

Student Marks Prediction using Machine Learning

1. Introduction

This project uses supervised machine learning to predict a student's final marks from study hours, attendance, assignment performance, and previous examination marks.

2. Objective

The objective is to build a simple regression model, evaluate its performance, and deploy it as a web application where a user can enter student details and receive a prediction.

3. Dataset

The project contains 120 student records. The target variable is final_marks. The dataset is included as student_marks.csv.

4. Data Preparation

The dataset is loaded with Pandas. The selected numeric features are used directly for modeling. The data is divided into training and testing sets using an 80:20 split.

5. Model Building

Linear Regression is used because the target variable, final marks, is numeric. The model is trained on the training data and then used to predict marks for unseen test data.

6. Model Evaluation

Metric	Result
Mean Squared Error (MSE)	21.47
R ² Score	0.722

7. Deployment

The trained model is connected to a Streamlit web interface. Users enter study hours, attendance, assignment score, and previous marks, then click Predict Marks to get the estimated final marks.

8. Files Included

app.py, train_model.py, Student_Marks_Prediction.ipynb, student_marks.csv, requirements.txt, README.md.

9. Conclusion

The project demonstrates the complete machine learning workflow: data preparation, train/test splitting, model training, evaluation, prediction, and deployment using Streamlit.